Orlaith Doyle

Orlaith Ní Dhúill

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Irish citizen, available immediately

Space scientist working with radar satellite data, with a focus on ocean applications. Recently an Earth observation intern at ESA ESTEC on the Biomass mission. Seeking roles in Earth observation, remote sensing and the wider space sector.

Experience

Earth Observation Intern, Biomass mission, European Space Agency, ESTEC, Netherlands — May to Oct 2026

- Developed and validated a sea surface salinity retrieval pipeline for Biomass P-band SAR; best scene within 0.4 psu of GLORYS12
- Built a resumable scanner cataloguing every Biomass acquisition (220,000+ products, ~5,000 unique locations)
- Identified a band-ordering mismatch in one product format that was swapping the HH and VV channels
- Prepared cal/val site files for the mission team; science and CAD contributions to the YPSat-2 payload

Founder and Tutor, Grade+ (physics and mathematics tutoring) — Feb to Oct 2024

Laboratory Technician, School of Physics, Dublin City University — Aug 2023 to Apr 2024

Education

MSc Space Science and Technology, University College Dublin. First Class Honours — 2025 to 2026

BSc (Hons) Physics with Astronomy, Dublin City University. Second Class Honours, Grade 1; third in class — 2019 to 2023

Leaving Certificate, Coláiste Íosagáin (all-Irish school). 510 CAO points — 2019

Selected projects

Pacific ocean front: polarimetric and coherence analysis of a Biomass scene, confirmed against GLORYS12 and OSTIA.

SQRT balloon: project manager for a five-person team, through PDR, CDR and flight acceptance to a stratospheric flight.

Blessington Lakes: SWOT water levels and Sentinel-2 analysis of Dublin's main reservoir.

EIRSAT-1 mission test: five live passes on the FlatSat in an operations crew, including safe-mode recovery.

TiNi mission concept: formation-flying gamma-ray mission to study Type Ia supernovae, from science case to spacecraft design.

BSc thesis: orbital modelling and launch windows to Jupiter using NASA JPL data.

Skills

Remote sensing: SAR processing, polarimetry (Pauli, H/α), InSAR, salinity retrieval and validation; SNAP, QGIS, ArcGIS; SMOS, SMAP, GLORYS12, ERA5, OSTIA, Sentinel-1/2/3, SWOT.

Space systems: spacecraft operations, mission and instrument design, systems engineering documentation, payload build and flight, radiation detectors, CAD.

Programming: Python (NumPy, pandas, xarray, SciPy, Astropy, Matplotlib, Plotly, Cartopy), MATLAB, Jupyter, Git, LaTeX, ESA MAAP and STAC, MicroPython.

Professional: project management, technical writing, presenting, teaching and outreach. Languages: English (C2), Irish (C2), French (B1), Dutch (A1). EU drone licence (A1/A3).

Outreach

Technical expert at ESA's CanSat event (2026); ESA school visits across Ireland (2026); youth coach, Blessington GAA (2016 to 2019); volunteer, Astronomy Ireland (2016 to 2018).

Reports, code and full project descriptions: orlaithdoyle.github.io